Thus $\frac{\phi_E}{c^2} = \frac{gR}{c^2}$;

Numerically, $\phi_E / c^2 = 6.96 \times 10^{-10}$ is very small. As above, (17) is approximately given by

$$T - T_0 \approx -\frac{\phi_E}{c^2} T_0 \ . \qquad (19)$$

Subtract (16), (19) to find satellite minus ground clock readings:

$$t - T \approx -\frac{\phi - \phi_E}{c^2} T_0 \ . \qquad (20)$$

Solve (19) for T_0 and substitute into (20):

$$t - T \approx -\frac{\phi - \phi_E}{c^2} / (1 - \frac{\phi_E}{c^2}) T \approx -\frac{\phi - \phi_E}{c^2} T = \frac{gR}{c^2} \frac{h}{R+h} T. \qquad (21)$$

The stated 24 is ground elapsed time T. Finally, after 24 , the satellite minus ground reading is

$$t - T = 46 \mu s \ . \qquad (22)$$

3.

1. For constant room temperature, the heat pump output must equal the room heat loss rate. Let q be the indoor heat output and P the electric input. The heat extracted outdoors at low temperature per unit time is $q-P$. Then

$$\frac{q-P}{P} \leqslant \frac{T_2}{T_1 - T_2} \ , \qquad (1)$$

T_1 is the indoor hot absolute temperature, T_2 the outdoor one. From (1),

$$P \geqslant \frac{T_1 - T_2}{T_1} q \ . \qquad (2)$$

Minimum cost requires minimum P: equality in " " in (2), the ideal case, minimizes P. Thus the least electric power is

$$P_{\min} = \frac{T_1 - T_2}{T_1} q \ . \qquad (3)$$

Heat conducts out of the room through the glass. The loss rate is

$$H = k \frac{T_1 - T_2}{l} S \ . \qquad (4)$$

For a constant room temperature,

$$q = H \ . \qquad (5)$$

Equations (3) ~ (5) give